

# Photosynthesis in Higher Plants

1. Which one of the following categories of organisms do not evolve oxygen during photosynthesis?

- (1) Red algae
- (2) Photosynthetic bacteria
- (3) C4 plants with Kranz anatomy
- (4) Blue green algae

2. Given below are two statements :

**Statement I:** The primary CO<sub>2</sub> acceptor in C4 plants is phosphoenolpyruvate and is found in the mesophyll cells.

**Statement II:** Mesophyll cells of C4 plants lack RuBisCo enzyme.

Choose the correct answer from the options given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Statement I is incorrect but Statement II is correct
- (4) Both Statement I and Statement II are correct

3. Where does cyclic photo-phosphorylation takes place?

- (1) membranous system of chloroplast
- (2) stroma lamellae
- (3) grana
- (4) stroma

4. The first dark reaction or CO<sub>2</sub> fixation product is the

- (1) 3C - Organic Acid
- (2) 6C - Organic Acid
- (3) 5C - Organic Acid
- (4) 1C - Organic Acid

5. ATP synthesis in chloroplast and mitochondria is due to proton gradient across the membrane. Select correct statement w.r.t. ATP formation in chloroplast

- (a) Proton accumulates in lumen of thylakoid
- (b) Splitting of water occurs on inner side of membrane
- (c) Proton accumulates in stroma side of chloroplast
- (d) NADP reductase is located on stroma side of membrane

- (1) Only (a) and (b) are correct
- (2) Only (b) and (c) are correct
- (3) Only (c) and (d) are correct
- (4) (a), (b) and (d) are correct

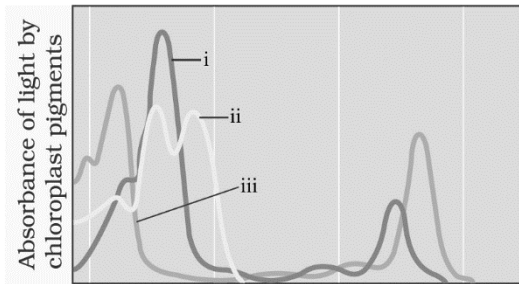
6. Which of the following statement is correct?

- (1) H<sub>2</sub>S is the hydrogen donor for purple and green sulphur bacteria
- (2) H<sub>2</sub>O is the hydrogen donor for purple and green sulphur bacteria
- (3) Oxygen is evolved by the purple and green sulphur bacteria
- (4) Oxygen evolved by the green plants comes from the CO<sub>2</sub>.

7. What does chemiosmosis requires?

- (1) membrane, cytosol, protonpump, ATPase and proton gradient
- (2) membrane, cytosol, protonpump and H-ions
- (3) membrane, protonpump, ATPase and proton gradient
- (4) membrane, cytosol, protonpump, ATPase and proton gradient and H-ions

8. Absorption spectrum of chlorophyll-'a' is shown by



- (1) i
- (2) ii
- (3) iii
- (4) Not shown

9. C<sub>3</sub> plants respond to higher CO<sub>2</sub> concentration by showing increased rates of photosynthesis because:

- (1) Current availability of CO<sub>2</sub> levels is limiting to the C<sub>3</sub> plants
- (2) C<sub>3</sub> plants show saturation at about 360  $\mu\text{L}^{-1}$  concentration of CO<sub>2</sub>
- (3) These plants responds to high CO<sub>2</sub> concentration even in the low light conditions
- (4) In these plants RuBisCO shows only carboxylation

10. Where does oxygen come from during photosynthesis?

- (1) Oxidation of water
- (2) Oxidation of glucose
- (3) Reduction of glucose
- (4) From CO<sub>2</sub> .

11. Cornelius van Niel, who, based on his studies demonstrated that photosynthesis is essentially a light-dependent reaction performed his experiments on:

- (1) Cladophora
- (2) purple and green bacteria
- (3) green algae
- (4) Non sulphur bacteria

12. Given below are two statements :

**Assertion – Cyclic photophosphorylation occurs in stroma**

**Reason – Stroma membrane lacks PS II and NADP reductase.**

**Choose correct answer-**

- (1) Both Assertion & Reason are correct and Reason is correct explanation for A
- (2) Both Assertion & Reason are correct but Reason is not correct explanation for Assertion
- (3) Assertion is correct but Reason is wrong
- (4) Both Assertion & Reason are wrong

13. The difference in development of proton gradient in photosynthesis and respiration is

- (1) Photosynthesis has thylakoid membrane across which proton gradient is generated
- (2) Proton accumulation is inside the membrane of thylakoid
- (3) both a and b
- (4) Mitochondrial membrane is involved

14. If only light of wavelength beyond 680 nm is available then which process will remain continue

- (1) cyclic photo-phosphorylation
- (2) non-cyclic photo-phosphorylation
- (3) ATP and NADPH
- (4) evolution of oxygen

15. Light is a limiting factor for plants in shade:

- (1) As they do not get full sunlight
- (2) As they do not get at most 10 % of full sunlight
- (3) As they do not get at least 50 % of full sunlight
- (4) As they do not get at most 70% full sunlight

16. At any point, the rate of photosynthesis is determined by the factor which is available at.....levels.

- (1) optimal
- (2) higher than required
- (3) sub-optimal
- (4) won't depend

**17. The C<sub>4</sub> plants grow very efficiently in warm climates because they**

- (1) Consume fewer ATP molecules for glucose synthesis
- (2) Minimize photorespiratory losses
- (3) Have more efficient chloroplasts
- (4) Contain enzymes which are not inactivated at high temperatures.

**18. The process which makes major difference between C<sub>3</sub> and C<sub>4</sub> Plants is**

- (1) lycosis
- (2) Clavin cycle
- (3) photorespiration
- (4) respiration

**19. Consider the following two statements:**

**I. Except for plants in shade or in dense forests, light is rarely a limiting factor in nature for photosynthesis.**

**II. Light saturation occurs at 50% of the full sunlight.**

- (1) Both statements are correct and II explains I
- (2) Both statements are incorrect
- (3) I is correct and II is incorrect
- (4) I is incorrect and II is correct

**20. The essential role of air in the growth of green plants was revealed by**

- (1) Priestley
- (2) Van Niel
- (3) Blackmann
- (4) Emerson

**21. Which of the following statements is not true regarding the C<sub>4</sub> plants?**

- (1) They show kranz anatomy
- (2) Decarboxylation process occurs in bundle sheath cells
- (3) Granal chloroplast is present in bundle sheath cells.
- (4) PEPcase enzyme activity occurs in mesophyll cells

**22. Respectively during the reaction of proton gradient and breakdown of proton gradient in chloroplast, protons move :-**

- (1) from the stroma to cytoplasm and from the cytoplasm to stroma
- (2) from the cytoplasm to stroma and from stroma to cytoplasm
- (3) from the thylakoid lumen to stroma and from the stroma to thylakoid lumen
- (4) from the stroma to thylakoid lumen and from the thylakoid lumen to stroma

**23. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).**

**Assertion – Chloroplasts usually align themselves along the walls of mesophyll cells.**

**Reason – They get optimum quantity of incident light by aligning along well.**

**In the light of the above statements, choose the correct answer from the options given below :**

- (1) Assertion and Reason are correct and Reason is correct explanation for Assertion
- (2) Assertion and Reason are correct but is not the explanation of Assertion
- (3) Assertion and Reason are both incorrect
- (4) Assertion is correct but Reason is incorrect

**24. The chloroplast stroma lamella membrane:**

- (1) Lack PS II as well as NADP reductase enzyme.
- (2) Lack PS II but have NADP reductase enzyme.
- (3) Lack NADP reductase enzyme but have PS II.
- (4) Have PS II as well as NADP reductase enzyme

**25. ATP and NADPH produced during photosynthesis, are used in which of the following reactions?**

- (1) Catabolic
- (2) Biosynthetic phase
- (3) amphibolic
- (4) Hydrolytic phase

**26. Which of the following statement is incorrect?**

- (1) Reaction centre is formed by a group of chl a molecules.
- (2) Reaction centre is different in both the photosystems.
- (3) In PS I the reaction centre chlorophyll a has an absorption peak at 700 nm.
- (4) In PS II the reaction centre chlorophyll a has absorption maxima at 680 nm.

**27. In C4 plants, photorespiration does not occur**

- (1) They have different RuBisCO.
- (2) They have no RuBisCO.
- (3) They have mechanism to increase the concentration of CO<sub>2</sub>
- (4) They cannot utilize ATP

**28. What is correct about C4 plant?**

- p. They have special type of leaf anatomy**
- q. They tolerate high temperature**
- r. They lack a process called photorespiration**
- s. Have lesser productivity**
- t. Show no response to highlight intensities**

- (1) pst
- (2) qrst
- (3) prst
- (4) pqr

**29. Given below are two statements:**

**Statement A – In CO fixation cycle, the molecules of ATP used is more than NADPH used.**

**Statement B – to meet the difference in number of ATP & NADPH used in dark reaction, cyclic phosphorylation take place.**

**In the light of the above statements, choose the correct answer from the options given below :**

- (1) A is correct but B is wrong
- (2) B is correct but A is wrong
- (3) A & B are correct and A explains B
- (4) A & B are incorrect.

**30. Photorespiration shows formation of**

- (1) sugar but not ATP
- (2) ATP but not sugar
- (3) both ATP and sugar
- (4) neither ATP nor sugar.

**31. The first stable product of Calvin cycle is**

- (1) 3-phosphoglycerate
- (2) 1, 3 biphosphoglycerate
- (3) glyceraldehyde - 3 phosphate
- (4) ribulose - 5- phosphate

**32. In Kranz anatomy, the bundle sheath cells have**

- (1) thin walls, many intercellular space and no chloroplast
- (2) thick walls, no intercellular spaces and large number of chloroplasts
- (3) thin walls, no intercellular spaces and several chloroplasts
- (4) thick walls, many intercellular space and few chloroplasts

**33. Which of the following is incorrect about the light reactions of photosynthesis?**

- (1) P680 and P700 are the reaction centres of PS-I, and PS-II respectively
- (2) Light energy provides energy for the photolysis of water through excitation of the reaction centre of PS-II
- (3) NADPH is not produced in cyclic electrons transport in light reactions
- (4) Reactions of the two photosystems are needed for the reduction of NADP

**34. Dark reactions of photosynthesis are actually not totally independent of light as:**

- (1) The initial reactions occur in the presence of light.
- (2) The reaction will be inhibited by the presence of light.
- (3) They can occur during the day as well.
- (4) They utilize the products of the light reaction.



**35. Read the following statements:**

- (a) F<sub>0</sub> part of ATPase is associated with breakdown of proton gradient.
- (b) A H-carrier contributes in creation of proton gradient.
- (c) Movement of electrons in ETS is coupled to pumping of protons into the lumen.
- (d) Formation of NADPH + H<sup>+</sup> is related with the creation of proton gradient.

**How many of the above statements are correct?**

**Select the correct option**

- (1) Two
- (2) One
- (3) Four
- (4) Three

**36. Which of the following statement is correct?**

- (1) The C<sub>3</sub> and C<sub>4</sub> plants respond similarly to CO<sub>2</sub> concentrations
- (2) At low light conditions C<sub>4</sub> plants show increase in the rates of photosynthesis
- (3) C<sub>4</sub> plants not respond to high temperature
- (4) The C<sub>3</sub> and C<sub>4</sub> plants respond differently to the CO<sub>2</sub> concentrations

**37. What is the correct ratio of ATP utilisation in steps of Calvin cycle?**

- (1) Reduction : Regeneration :: 1 : 1
- (2) Reduction : Regeneration :: 2 : 1
- (3) Reduction : Regeneration :: 2 : 2
- (4) Reduction : Regeneration :: 1 : 2

**38. The C<sub>4</sub> – plants are photosynthetically more efficient than C<sub>3</sub> – plants because**

- (1) The CO<sub>2</sub> compensation point is more
- (2) CO<sub>2</sub> generated during photorespiration is trapped and recycled through PEP carboxylase
- (3) The CO<sub>2</sub> efflux is not prevented
- (4) They have more chloroplasts.

**39. Identify the correct statements regarding chemiosmotic hypothesis :**

- (a) Splitting of the water molecule takes place on the inner side of the membrane.
- (b) Protons accumulate within the lumen of the thylakoids.
- (c) Primary acceptor of electron transfers the electrons to an electron carrier.
- (d) NADP reductase enzyme is located on the stroma side of the membrane.
- (e) Protons increase in number in stroma.

**Choose the correct answer from the options given below:**

- (1) (a), (b) and (e)
- (2) (a), (b) and (d)
- (3) (b), (c) and (d)
- (4) (b), (c) and (e)

**40. For assimilation of one CO<sub>2</sub> molecule via photosynthesis, the energy required in form of ATP and NADPH<sub>2</sub> are:**

- (1) 2 ATP and 2 NADPH<sub>2</sub>
- (2) 5 ATP and 3 NADPH<sub>2</sub>
- (3) 3 ATP and 2 NADPH<sub>2</sub>
- (4) 18 ATP and 12 NADPH<sub>2</sub>

**41. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?**

- (1) Light saturation for CO<sub>2</sub> -fixation occurs at 10% of full sunlight.
- (2) Increasing atmospheric CO<sub>2</sub> concentration upto 0.05% can enhance the CO<sub>2</sub> -fixation rate
- (3) C<sub>3</sub> plants respond to higher temperature with enhanced photosynthesis, while C<sub>4</sub> plants have a much lower temperature optimum.
- (4) Tomato is a greenhouse crop, which can be grown in a CO<sub>2</sub> -enriched atmosphere for higher yield.

42. Given below are two statements:

**Assertion – Chlorophyll 'a' is the chief pigment associated with photosynthesis**

**Reason – Chlorophyll maximum absorption coincides with maximum photosynthesis.**

**Choose the correct answer from the options given below:**

- (1) Assertion & Reason are correct & Reason is correct explanation of Assertion
- (2) Assertion & Reason are correct but Reason is not correct explanation of Assertion
- (3) Assertion is correct & Reason is incorrect.
- (4) Assertion & Reason are incorrect.

43. Which of the following statements are correct regarding synthesis of ATP in chloroplast during photosynthesis ?

**(A) Splitting of water in stroma helps in creation of proton gradient**

**(B) Cytochrome complex helps in the release of protons in the lumen of thylakoid by accepting electrons from hydrogen carrier.**

**(C) Movement of protons across the membrane to the stroma through the  $F_0$  of the ATPase is coupled with ATP synthesis.**

**(D) Reduction of  $NADP^+$  to  $NADPH + H^+$  is also a cause for creation of proton gradient.**

**Select the correct option**

- (1) All statements are correct
- (2) C and D
- (3) A and B
- (4) B, C and D

44. Choose the correct sequence of steps involved in  $C_4$  cycle.

- (1)  $CO_2$  fixation  $\rightarrow$  Regeneration  $\rightarrow$  Transport
- (2)  $CO_2$  fixation  $\rightarrow$  Recarboxylation  $\rightarrow$  Decarboxylation
- (3) Transport  $\rightarrow$  Regeneration  $\rightarrow$   $CO_2$  fixation
- (4)  $CO_2$  fixation  $\rightarrow$  Transport  $\rightarrow$  Decarboxylation

45. Given below are two statements :

**Statement A – The accumulation of protons in stroma cause decrease in pH of stroma.**

**Statement B – The accumulation of proton in lumen cause formation of potential gradient across thylakoid membrane.**

**Select the correct option-**

- (1) A is correct and B is incorrect
- (2) B is correct and A is incorrect
- (3) A & B are correct
- (4) A & B are incorrect

46. Which of the following statements is correct in relation to the photosynthesis ?

- (1) It is the redox reaction during which reduction of  $H_2O$  and oxidation of  $CO_2$  occurs
- (2) First of all it started in Chlorella
- (3) It converts light energy into chemical energy in night
- (4) Photosynthesis results in synthesis of sugar

47. Before the discovery of RuBP as primary acceptor of  $CO_2$  in Calvin cycle, it was believed that the primary acceptor would be :-

- (1) a 6-carbon compound
- (2) a 2-carbon compound
- (3) a 3-carbon compound
- (4) a 4-carbon compound

48. Given below are two statements;

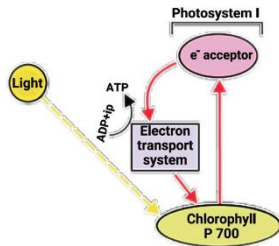
**Statement A – Biosynthetic reaction is independent of direct presence of light**

**Statement B – Biosynthetic process continues for some time after the light becomes unavailable and then stops.**

**Choose the correct answer from the option given below :**

- (1) Both A & B are correct
- (2) A is correct and B is incorrect
- (3) A is incorrect and B is correct
- (4) A & B are in correct

49. What would be true for the events shown in the given diagram?



- I. A possible location where this could be happening is in the stroma lamellae.  
 II. While the membrane or lamellae of the grana have both PS I and PS II the stroma lamellae membranes lack PS II as well as NADP reductase enzyme.

- (1) Only I  
 (2) Only II  
 (3) Both I and II  
 (4) Neither I nor II

50. In Hatch and Slack pathway:

- I. The C acid OAA is formed in the mesophyll cells.  
 II. Malic acid or aspartic acid are transported to the bundle sheath cells.  
 III. C acids are broken down to release carbon dioxide and a 3 carbon molecule in mesophyll cells.  
 IV. The 3 carbon molecule is transported back to the bundle sheath cell where it is converted to PEP again.

- (1) Only I and II are correct  
 (2) II, III and IV are incorrect  
 (3) Only I, II and III are correct  
 (4) I, II, III and IV are correct

## Answer Key

1. (2)
2. (4)
3. (2)
4. (1)
5. (4)
6. (1)
7. (3)
8. (3)
9. (1)
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47. (2)
48. (1)
49. (3)
50. (1)

